

Anomaly Detection with Autoencoders

Lecture Note — Module 2, Blocks 1–2

MS Data Engineering

Contents

1	Introduction and Motivation	2
2	The Autoencoder Framework	2
2.1	The Bottleneck as an Inductive Bias	3
2.2	Architecture Variants	3
3	The Vanilla (Deterministic) Autoencoder	4
3.1	Architecture	4
3.2	Semi-Supervised Training Protocol	4
4	Anomaly Scoring and Threshold Selection	5
4.1	Percentile-Based Thresholds	5
4.2	Statistical Thresholds	5
4.3	Extreme Value Theory (EVT)	5
4.4	Supervised Threshold Selection	6
5	Evaluation Metrics for Anomaly Detection	6
6	The Bottleneck Dimension: Bias–Variance Trade-Off	7
7	Variational Autoencoders (VAE)	8
7.1	Probabilistic Formulation	8
7.2	The Evidence Lower Bound (ELBO)	9
7.3	The Reparameterisation Trick	9
7.4	Closed-Form KL Divergence	10
7.5	VAE Loss Function	10
7.6	VAE Anomaly Score	11
7.7	Reconstruction Probability	11
7.8	The β -VAE and Disentanglement	12
8	Latent Space Geometry and Interpretation	12
8.1	Vanilla AE Latent Space	12
8.2	VAE Latent Space	12
9	Comparison: Vanilla AE vs VAE	13
10	Limitations and Extensions	13
10.1	Limitations of Autoencoder-Based Anomaly Detection	13
10.2	Extensions	13

Table 1: *

	Symbol	Meaning
Notation	$\mathbf{x} \in \mathbb{R}^{d_x}$	Input observation
	$\hat{\mathbf{x}}$	Reconstructed output
	$\mathbf{z} \in \mathbb{R}^{d_z}$	Latent code / embedding
	f_θ	Encoder with parameters θ
	g_ϕ	Decoder with parameters ϕ
	d_x, d_z	Input and latent dimensionality
	$s(\mathbf{x})$	Anomaly score
	τ	Anomaly threshold
	$\boldsymbol{\mu}_\theta, \boldsymbol{\sigma}_\theta$	VAE encoder outputs (mean, std. dev.)
	$q_\theta(\mathbf{z} \mathbf{x})$	Approximate posterior (encoder)
	$p_\phi(\mathbf{x} \mathbf{z})$	Likelihood (decoder)
	$p(\mathbf{z})$	Prior over latent space
	$\text{KL}(\cdot \cdot)$	Kullback–Leibler divergence
	β	KL weight in β -VAE
	$\odot$	Element-wise (Hadamard) product

1 Introduction and Motivation

Anomaly detection—also known as outlier detection or novelty detection—is the task of identifying observations that deviate significantly from the majority of the data. It arises naturally in domains such as network intrusion detection, industrial fault monitoring, financial fraud prevention, and medical diagnostics. In all of these settings, the key challenge is the same: anomalies are *rare*, *diverse*, and often impossible to enumerate in advance.

Classical approaches include statistical methods (e.g. Grubbs’ test, Mahalanobis distance), density-based methods (e.g. kernel density estimation, Local Outlier Factor), and tree-based methods (e.g. Isolation Forest). While effective on low-to-moderate dimensional data, these methods often struggle with the high-dimensional, nonlinear manifolds that characterise modern datasets.

Autoencoders offer a compelling deep-learning-based alternative. By learning to compress and reconstruct data through a low-dimensional bottleneck, an autoencoder implicitly learns a model of *normality*. Observations that cannot be faithfully reconstructed are flagged as anomalous. This simple yet powerful principle is the foundation of this lecture.

2 The Autoencoder Framework

Definition 2.1: Autoencoder

An **autoencoder** is a neural network composed of two sub-networks:

- An **encoder** $f_\theta : \mathbb{R}^{d_x} \rightarrow \mathbb{R}^{d_z}$, mapping an input $\mathbf{x}$ to a latent representation $\mathbf{z} = f_\theta(\mathbf{x})$.
- A **decoder** $g_\phi : \mathbb{R}^{d_z} \rightarrow \mathbb{R}^{d_x}$, producing a reconstruction $\hat{\mathbf{x}} = g_\phi(\mathbf{z})$.

The latent space $\mathbb{R}^{d_z}$ satisfies $d_z < d_x$ and is called the **bottleneck**. The vector $\mathbf{z}$ is referred to as the **latent code** or **embedding**.

Definition 2.2: Reconstruction Loss

The **reconstruction loss** of an autoencoder on a dataset $\{\mathbf{x}_1, \dots, \mathbf{x}_N\}$ is

$$\mathcal{L}_{\text{AE}} = \frac{1}{N} \sum_{i=1}^N \|\mathbf{x}_i - \hat{\mathbf{x}}_i\|^2, \quad (1)$$

where $\hat{\mathbf{x}}_i = g_\phi(f_\theta(\mathbf{x}_i))$. The most common instantiation uses the mean squared error (MSE); alternatives include binary cross-entropy when the data lies in $[0, 1]^{d_x}$.

The autoencoder is trained end-to-end by minimising $\mathcal{L}_{\text{AE}}$ over the parameters (θ, ϕ) using stochastic gradient descent or one of its variants (Adam, RMSProp, etc.). The constraint $d_z < d_x$ forces the network to learn a compressed, lower-dimensional representation of the input data.

2.1 The Bottleneck as an Inductive Bias

The reduced dimensionality of the latent space acts as an *information bottleneck*. During training on normal data only, the encoder learns to project inputs onto a low-dimensional manifold that captures the principal modes of variation of the normal distribution. The decoder, in turn, learns the inverse mapping from this manifold back to input space.

Proposition 2.1: Anomaly Detection Principle

Let f_θ, g_ϕ be an autoencoder trained on a dataset of normal observations drawn from a distribution P_{normal} . Then for a new observation $\mathbf{x}$:

1. If $\mathbf{x} \sim P_{\text{normal}}$, the reconstruction $\hat{\mathbf{x}} = g_\phi(f_\theta(\mathbf{x}))$ is expected to be close to $\mathbf{x}$, yielding a **small** reconstruction error.
2. If $\mathbf{x} \not\sim P_{\text{normal}}$ (i.e. $\mathbf{x}$ is anomalous), the bottleneck prevents faithful compression, resulting in a **large** reconstruction error.

Definition 2.3: Anomaly Score

The **anomaly score** of an observation $\mathbf{x}$ under an autoencoder (f_θ, g_ϕ) is

$$s(\mathbf{x}) = \|\mathbf{x} - g_\phi(f_\theta(\mathbf{x}))\|^2. \quad (2)$$

A sample is classified as anomalous if $s(\mathbf{x}) > \tau$ for some threshold $\tau > 0$.

2.2 Architecture Variants

The autoencoder framework admits several architectural variants, summarised in Table 2.

Table 2: Common autoencoder variants for anomaly detection.

Variant	Encoder/Decoder	Bottleneck	Typical use
Vanilla AE	Dense MLP	Deterministic vector $\mathbf{z}$	Tabular data
Conv AE	CNN	Spatial feature map	Images
LSTM-AE	LSTM	Hidden state sequence	Time series
VAE	MLP + reparam.	Stochastic $\mathbf{z} \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\sigma}^2)$	Generative + detection

3 The Vanilla (Deterministic) Autoencoder

The simplest form of autoencoder uses fully connected (dense) layers with nonlinear activation functions. For tabular data of dimension d_x , a typical architecture consists of an encoder with layers of decreasing width and a decoder that mirrors this structure symmetrically.

3.1 Architecture

A vanilla autoencoder for d_x -dimensional input typically takes the form:

$$\text{Encoder: } \mathbf{h}_1 = \sigma(\mathbf{W}_1 \mathbf{x} + \mathbf{b}_1), \quad \mathbf{h}_2 = \sigma(\mathbf{W}_2 \mathbf{h}_1 + \mathbf{b}_2), \quad \mathbf{z} = \mathbf{W}_3 \mathbf{h}_2 + \mathbf{b}_3, \quad (3)$$

$$\text{Decoder: } \mathbf{h}'_2 = \sigma(\mathbf{W}'_2 \mathbf{z} + \mathbf{b}'_2), \quad \mathbf{h}'_1 = \sigma(\mathbf{W}'_1 \mathbf{h}'_2 + \mathbf{b}'_1), \quad \hat{\mathbf{x}} = \mathbf{W}'_0 \mathbf{h}'_1 + \mathbf{b}'_0, \quad (4)$$

where σ denotes a nonlinear activation (typically ReLU). The weight matrices and biases $\{\mathbf{W}_k, \mathbf{b}_k, \mathbf{W}'_k, \mathbf{b}'_k\}$ constitute the trainable parameters θ and ϕ respectively.

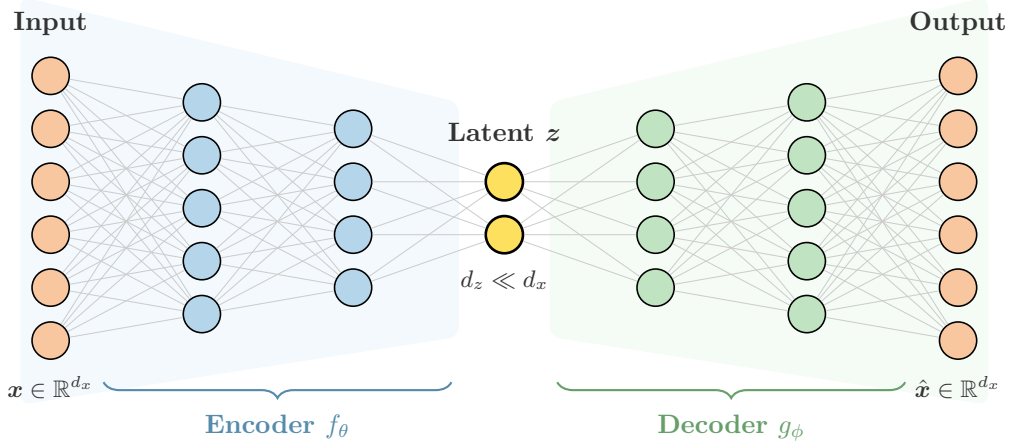


Figure 1: Architecture of a vanilla autoencoder. The input is progressively compressed through the encoding layers to a low-dimensional latent code $\mathbf{z}$ (bottleneck), then reconstructed by the symmetric decoder.

3.2 Semi-Supervised Training Protocol

A critical distinction from standard autoencoder usage is the *semi-supervised training protocol*: the autoencoder is trained exclusively on normal (non-anomalous) data. This ensures that the learned manifold corresponds to the structure of normality. At inference time, both normal and potentially anomalous data are passed through the model, and anomaly scores are computed via Eq. (2).

Remark 3.1: Connection to PCA

A linear autoencoder (i.e. with no activation functions) with MSE loss learns the same subspace as Principal Component Analysis (PCA). Specifically, the columns of the encoder's weight matrix span the subspace of the top d_z principal components of the data covariance matrix. The nonlinear autoencoder can therefore be viewed as a **nonlinear generalisation of PCA**.

Theorem 3.1: Linear AE-PCA Equivalence

Let $\mathbf{X} \in \mathbb{R}^{N \times d_x}$ be a centred data matrix (zero column means) and consider a linear autoencoder with encoder $\mathbf{z} = \mathbf{W}_e \mathbf{x}$, $\mathbf{W}_e \in \mathbb{R}^{d_z \times d_x}$, and decoder $\hat{\mathbf{x}} = \mathbf{W}_d \mathbf{z}$, $\mathbf{W}_d \in \mathbb{R}^{d_x \times d_z}$. The MSE-optimal solution satisfies:

1. The column space of $\mathbf{W}_d^*$ equals the subspace spanned by the top d_z eigenvectors of $\mathbf{\Sigma} = \frac{1}{N} \mathbf{X}^\top \mathbf{X}$.
2. The minimum reconstruction error equals $\sum_{j=d_z+1}^{d_x} \lambda_j$, where $\lambda_1 \geq \dots \geq \lambda_{d_x}$ are the eigenvalues of $\mathbf{\Sigma}$.

Proof. The reconstruction error is $\mathcal{L} = \frac{1}{N} \sum_{i=1}^N \|\mathbf{x}_i - \mathbf{W}_d \mathbf{W}_e^\top \mathbf{x}_i\|^2 = \frac{1}{N} \|\mathbf{X} - \mathbf{X} \mathbf{W}_e^\top \mathbf{W}_d^\top\|_F^2$. The product $\mathbf{P} = \mathbf{W}_d \mathbf{W}_e^\top$ is a rank- d_z linear map. By the Eckart–Young–Mirsky theorem, the rank- d_z matrix that minimises $\|\mathbf{X} - \mathbf{X} \mathbf{P}^\top\|_F^2$ is the orthogonal projection onto the top d_z eigenvectors of $\mathbf{X}^\top \mathbf{X}$. The residual is $\sum_{j>d_z} \lambda_j$. $\square$

Remark 3.2: From linear to nonlinear: universal approximation

Theorem 3.1 shows that a linear bottleneck can only capture principal subspaces. By introducing nonlinear activations (ReLU, sigmoid, etc.), the autoencoder gains the capacity to approximate *nonlinear* manifolds. The universal approximation theorem for neural networks guarantees that, given sufficient width, a single-hidden-layer encoder can approximate any continuous mapping $f : \mathbb{R}^{d_x} \rightarrow \mathbb{R}^{d_z}$ to arbitrary precision on compact domains. In practice, deeper architectures with multiple hidden layers achieve this with far fewer parameters, motivating the multi-layer designs used in modern autoencoders.

4 Anomaly Scoring and Threshold Selection

Once the autoencoder is trained, the reconstruction error $s(\mathbf{x})$ serves as the anomaly score. The remaining challenge is to select a threshold τ that separates normal from anomalous observations.

4.1 Percentile-Based Thresholds

Compute the reconstruction errors on the training set (which contains only normal data) and set

$$\tau = Q_p(\{s(\mathbf{x}_i)\}_{i=1}^N), \quad (5)$$

where Q_p denotes the p -th percentile. Common choices are $p = 95$ or $p = 99$. This approach assumes that the top $(100 - p)\%$ of training errors represent the tail of the normal distribution, and any test sample exceeding this threshold is anomalous.

4.2 Statistical Thresholds

An alternative is the mean-plus- k -sigma rule:

$$\tau = \mu_{\text{train}} + k \cdot \sigma_{\text{train}}, \quad (6)$$

where μ_{train} and σ_{train} are the mean and standard deviation of training reconstruction errors, and k is a sensitivity parameter (typically $k \in \{2, 3\}$). This assumes approximate normality of the score distribution.

4.3 Extreme Value Theory (EVT)

A more principled approach models the tail of the training score distribution using Extreme Value Theory. By the Pickands–Balkema–de Haan theorem, exceedances above a sufficiently high threshold follow a *Generalised Pareto Distribution* (GPD).

Theorem 4.1: Pickands–Balkema–de Haan (EVT)

Let F be the distribution function of the anomaly scores on normal data. For a sufficiently high threshold u , the conditional excess distribution

$$F_u(y) = P(s - u \leq y \mid s > u), \quad y \geq 0,$$

is well approximated by a Generalised Pareto Distribution $\text{GPD}(\xi, \beta)$ with shape parameter ξ and scale parameter $\beta > 0$:

$$G_{\xi, \beta}(y) = \begin{cases} 1 - (1 + \xi y / \beta)^{-1/\xi} & \text{if } \xi \neq 0, \\ 1 - e^{-y/\beta} & \text{if } \xi = 0. \end{cases}$$

Fitting the GPD to the tail of training scores provides a statistically grounded anomaly threshold with an associated false-positive rate.

4.4 Supervised Threshold Selection

When a labelled validation set is available, the threshold can be chosen to maximise a metric of interest (e.g. the F_1 score) by grid search over candidate thresholds.

5 Evaluation Metrics for Anomaly Detection

Anomaly detection operates in a heavily imbalanced setting: the anomaly class is typically a small minority. Standard accuracy is therefore a poor metric. The following metrics are preferred.

Definition 5.1: ROC Curve and AUC-ROC

The **Receiver Operating Characteristic (ROC) curve** plots the True Positive Rate (TPR) against the False Positive Rate (FPR) as the threshold τ varies:

$$\text{TPR}(\tau) = \frac{\text{TP}(\tau)}{\text{TP}(\tau) + \text{FN}(\tau)}, \quad \text{FPR}(\tau) = \frac{\text{FP}(\tau)}{\text{FP}(\tau) + \text{TN}(\tau)}.$$

The **Area Under the ROC Curve (AUC-ROC)** summarises classification performance across all thresholds. $\text{AUC} = 1$ indicates perfect separation; $\text{AUC} = 0.5$ indicates random guessing.

Definition 5.2: Precision–Recall Curve and Average Precision

The **Precision–Recall (PR) curve** plots Precision against Recall as τ varies:

$$\text{Precision}(\tau) = \frac{\text{TP}(\tau)}{\text{TP}(\tau) + \text{FP}(\tau)}, \quad \text{Recall}(\tau) = \frac{\text{TP}(\tau)}{\text{TP}(\tau) + \text{FN}(\tau)}.$$

The **Average Precision (AP)** is the area under the PR curve. In highly imbalanced settings, the PR curve is often more informative than the ROC curve because it focuses on the minority (anomalous) class.

Remark 5.1: Why PR is preferred under class imbalance

When the anomaly rate is very low (e.g. 1%), a classifier that labels everything as normal achieves $\text{FPR} \approx 0$, making the ROC curve appear favourable. The PR curve penalises such behaviour because precision drops when the model fails to detect anomalies without incurring

many false positives.

Definition 5.3: F_1 Score

The F_1 score is the harmonic mean of precision and recall:

$$F_1 = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}.$$

It provides a single number that balances both concerns and is commonly used to select an operating threshold when labelled validation data are available.

Theorem 5.1: Probabilistic Interpretation of AUC-ROC

Let $s(\mathbf{x})$ be a scoring function (e.g. reconstruction error) and let $X^+ \sim P_{\text{anomaly}}$, $X^- \sim P_{\text{normal}}$ be drawn independently. Then:

$$\text{AUC} = P(s(X^+) > s(X^-)). \quad (7)$$

That is, the AUC equals the probability that a randomly chosen anomalous sample receives a higher anomaly score than a randomly chosen normal sample.

Proof. By definition, $\text{AUC} = \int_0^1 \text{TPR}(\text{FPR}^{-1}(u)) du$. Substituting the definitions of TPR and FPR as survival functions of the score distributions F^+ and F^- and changing variables yields:

$$\text{AUC} = \int_{-\infty}^{\infty} F^-(t) dF^+(t) = P(s(X^+) > s(X^-)),$$

where the last equality uses the definition of the convolution of two CDFs (the Wilcoxon–Mann–Whitney statistic). $\square$

Proposition 5.1: Base-Rate Fallacy in Anomaly Detection

Consider an anomaly detector with sensitivity $\text{TPR} = 0.95$ and specificity $1 - \text{FPR} = 0.99$ applied to a population with anomaly prevalence $\pi = 0.001$ (i.e. 1 in 1000). The **positive predictive value** (PPV)—the probability that a flagged sample is truly anomalous—is:

$$\text{PPV} = \frac{\text{TPR} \cdot \pi}{\text{TPR} \cdot \pi + \text{FPR} \cdot (1 - \pi)} = \frac{0.95 \times 0.001}{0.95 \times 0.001 + 0.01 \times 0.999} \approx 0.087.$$

Even with excellent TPR and FPR, the vast majority ($\sim 91\%$) of flagged samples are false positives. This **base-rate fallacy** is a fundamental challenge in anomaly detection and underscores the importance of precision-focused metrics.

6 The Bottleneck Dimension: Bias–Variance Trade-Off

The latent dimension d_z is the most critical hyperparameter of an autoencoder for anomaly detection. It controls the expressiveness of the learned representation and directly affects the model’s ability to discriminate between normal and anomalous data.

Theorem 6.1: Bottleneck Trade-Off (Informal)

Let (f_θ, g_ϕ) be an autoencoder with latent dimension d_z , trained on normal data drawn from a distribution supported on a manifold $\mathcal{M}$ of intrinsic dimension $d_{\mathcal{M}}$.

1. **Underfitting** ($d_z \ll d_{\mathcal{M}}$): the autoencoder cannot capture the full structure of $\mathcal{M}$, and even normal samples incur high reconstruction error. Discrimination between normal

and anomalous samples is poor.

2. **Overfitting** ($d_z \gg d_M$): the autoencoder has enough capacity to reconstruct even anomalous inputs through the bottleneck, again yielding poor discrimination.
3. **Optimal regime** ($d_z \approx d_M$): the model has enough capacity to faithfully encode normal patterns while being sufficiently constrained to reject anomalies.

Remark 6.1: Practical guidance

In practice, d_z is selected by cross-validation or by monitoring AUC-ROC on a validation set containing a small number of labelled anomalies. A sweep over candidate values (e.g. $d_z \in \{2, 4, 8, 12, 16\}$) typically reveals a peaked curve, with performance degrading at both extremes.

7 Variational Autoencoders (VAE)

The Variational Autoencoder (VAE), introduced by Kingma and Welling [1], extends the deterministic autoencoder by placing a probabilistic structure on the latent space. Rather than encoding each input to a single point, the VAE encodes it as a *distribution* over latent codes.

7.1 Probabilistic Formulation

Definition 7.1: Variational Autoencoder

A **Variational Autoencoder** consists of:

1. An **encoder** (inference network) $q_\theta(z | x) = \mathcal{N}(\mu_\theta(x), \text{diag}(\sigma_\theta^2(x)))$ that maps an input x to the parameters of a Gaussian distribution over latent codes.
2. A **decoder** (generative network) $p_\phi(x | z)$ that maps a sampled latent code z to a distribution over reconstructions.
3. A **prior** $p(z) = \mathcal{N}(0, I)$ over the latent space.

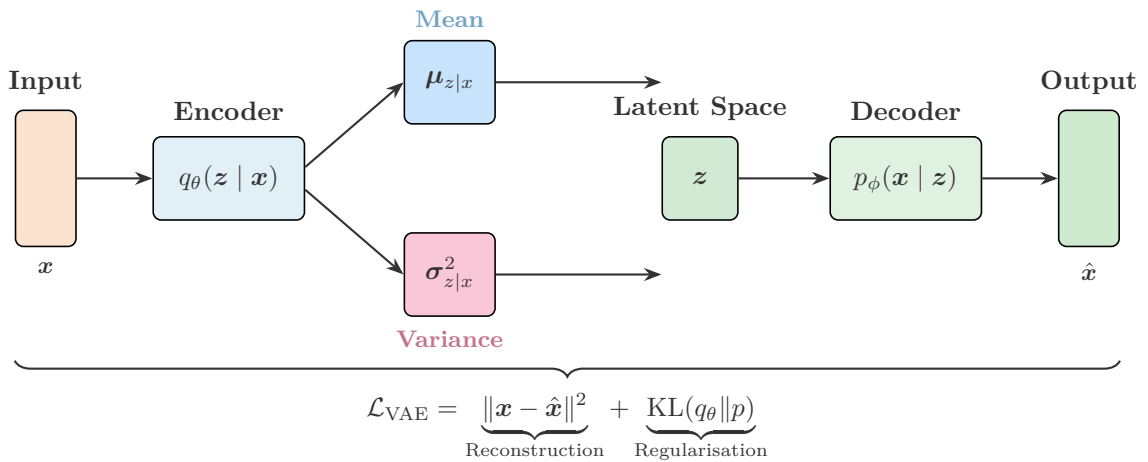


Figure 2: Architecture of a Variational Autoencoder. The encoder outputs the mean μ and variance σ^2 of the approximate posterior. A latent code z is sampled via the reparameterisation trick and passed to the decoder. The loss combines reconstruction quality with KL regularisation.

7.2 The Evidence Lower Bound (ELBO)

The VAE is trained by maximising a lower bound on the log-evidence $\log p(\mathbf{x})$, known as the Evidence Lower Bound.

Theorem 7.1: ELBO Decomposition

For any observation $\mathbf{x}$, the marginal log-likelihood decomposes as

$$\log p(\mathbf{x}) = \underbrace{\mathbb{E}_{q_\theta(\mathbf{z}|\mathbf{x})}[\log p_\phi(\mathbf{x}|\mathbf{z})] - \text{KL}(q_\theta(\mathbf{z}|\mathbf{x})\|p(\mathbf{z}))}_{\text{ELBO}(\mathbf{x})} + \text{KL}(q_\theta(\mathbf{z}|\mathbf{x})\|p(\mathbf{z}|\mathbf{x})). \quad (8)$$

Since $\text{KL}(q_\theta(\mathbf{z}|\mathbf{x})\|p(\mathbf{z}|\mathbf{x})) \geq 0$, the ELBO is a lower bound on $\log p(\mathbf{x})$. Maximising the ELBO jointly improves reconstruction quality (first term) and encourages the approximate posterior to match the prior (second term).

Proof. By Bayes' rule, $p(\mathbf{z}|\mathbf{x}) = p(\mathbf{x}|\mathbf{z})p(\mathbf{z})/p(\mathbf{x})$. Taking the expectation of $\log p(\mathbf{x})$ under $q_\theta(\mathbf{z}|\mathbf{x})$ and rearranging:

$$\begin{aligned} \log p(\mathbf{x}) &= \mathbb{E}_q[\log p(\mathbf{x})] = \mathbb{E}_q\left[\log \frac{p(\mathbf{x}, \mathbf{z})}{p(\mathbf{z}|\mathbf{x})}\right] \\ &= \mathbb{E}_q\left[\log \frac{p(\mathbf{x}, \mathbf{z})}{q_\theta(\mathbf{z}|\mathbf{x})}\right] + \mathbb{E}_q\left[\log \frac{q_\theta(\mathbf{z}|\mathbf{x})}{p(\mathbf{z}|\mathbf{x})}\right] \\ &= \text{ELBO}(\mathbf{x}) + \text{KL}(q_\theta(\mathbf{z}|\mathbf{x})\|p(\mathbf{z}|\mathbf{x})). \end{aligned}$$

Expanding the joint $p(\mathbf{x}, \mathbf{z}) = p_\phi(\mathbf{x}|\mathbf{z})p(\mathbf{z})$ in the ELBO term gives the decomposition in Eq. (8). $\square$

7.3 The Reparameterisation Trick

Sampling $\mathbf{z} \sim q_\theta(\mathbf{z}|\mathbf{x})$ introduces a stochastic node into the computation graph, which prevents direct backpropagation. The *reparameterisation trick* resolves this by expressing the sample as a deterministic function of the encoder outputs and an independent noise variable:

$$\mathbf{z} = \boldsymbol{\mu}_\theta(\mathbf{x}) + \boldsymbol{\epsilon} \odot \boldsymbol{\sigma}_\theta(\mathbf{x}), \quad \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}). \quad (9)$$

This transforms the stochastic sampling into a deterministic transformation with external noise, allowing gradients to flow through $\boldsymbol{\mu}$ and $\boldsymbol{\sigma}$ during backpropagation.

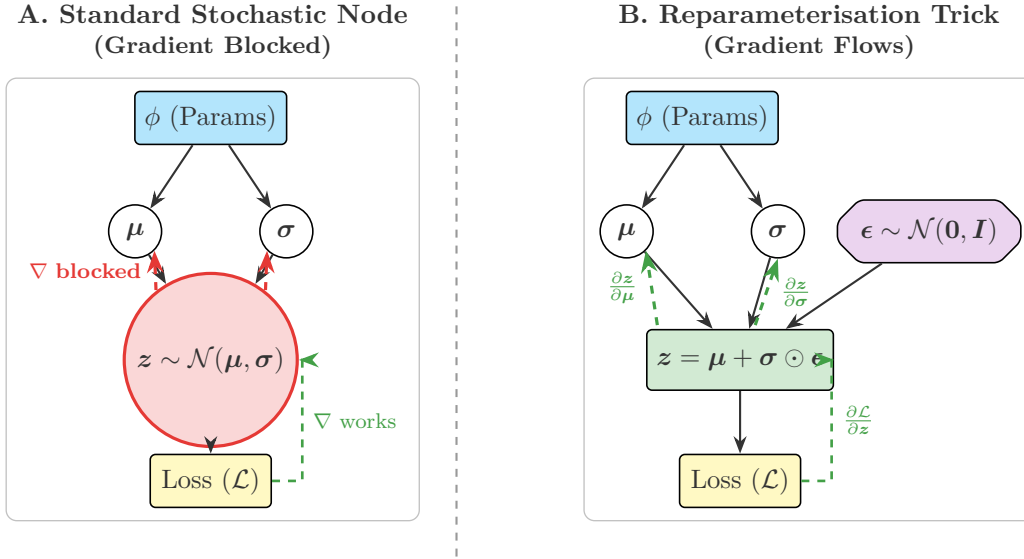


Figure 3: The reparameterisation trick. **(A)** Standard stochastic sampling $z \sim \mathcal{N}(\mu, \sigma)$ blocks gradients through the sampling node. **(B)** The reparameterisation trick moves stochasticity to an external input $\epsilon \sim \mathcal{N}(0, I)$ and expresses z as a deterministic function $z = \mu + \sigma \odot \epsilon$, allowing gradients to flow to the encoder parameters ϕ .

7.4 Closed-Form KL Divergence

When both the approximate posterior and the prior are Gaussian, the KL divergence admits a closed-form expression.

Proposition 7.1: KL Divergence for Diagonal Gaussians

Let $q = \mathcal{N}(\mu, \text{diag}(\sigma^2))$ and $p = \mathcal{N}(0, I)$, both in $\mathbb{R}^d$. Then

$$\text{KL}(q||p) = \frac{1}{2} \sum_{j=1}^d [\mu_j^2 + \sigma_j^2 - \log \sigma_j^2 - 1]. \quad (10)$$

Proof. By definition,

$$\text{KL}(q||p) = \mathbb{E}_q \left[\log \frac{q(z)}{p(z)} \right].$$

Since both distributions are diagonal Gaussians, the KL decomposes into a sum over independent dimensions. For a single dimension j :

$$\begin{aligned} \text{KL}(\mathcal{N}(\mu_j, \sigma_j^2) || \mathcal{N}(0, 1)) &= \mathbb{E} \left[\log \frac{\sigma_j^{-1} \exp(-\frac{(z_j - \mu_j)^2}{2\sigma_j^2})}{\exp(-\frac{z_j^2}{2})} \right] \\ &= -\frac{1}{2} \log \sigma_j^2 - \frac{1}{2} + \frac{1}{2} (\mu_j^2 + \sigma_j^2). \end{aligned}$$

Summing over $j = 1, \dots, d$ yields Eq. (10). □

7.5 VAE Loss Function

In practice, the negative ELBO is minimised as a loss function. For a Gaussian decoder $p_\phi(\mathbf{x} | z) = \mathcal{N}(g_\phi(z), I)$, the reconstruction term reduces to MSE, giving:

$$\mathcal{L}_{\text{VAE}} = \|\mathbf{x} - \hat{\mathbf{x}}\|^2 + \beta \cdot \text{KL}(q_\theta(z | \mathbf{x}) || p(z)), \quad (11)$$

where $\beta = 1$ recovers the standard ELBO and $\beta > 1$ gives the β -VAE variant [2], which encourages a more disentangled and compact latent space at the cost of reconstruction fidelity.

7.6 VAE Anomaly Score

Definition 7.2: ELBO-Based Anomaly Score

The anomaly score of an observation $\mathbf{x}$ under a trained VAE is

$$s_{\text{VAE}}(\mathbf{x}) = \|\mathbf{x} - \hat{\mathbf{x}}\|^2 + \text{KL}(q_{\theta}(\mathbf{z} | \mathbf{x}) \| p(\mathbf{z})). \quad (12)$$

This score captures two complementary signals: (i) poor reconstruction quality, and (ii) deviation of the encoded distribution from the expected prior. An anomaly may trigger one or both components.

Remark 7.1: Multiple sampling for score stability

Because the VAE's reconstruction involves a stochastic sample $\mathbf{z} \sim q(\mathbf{z} | \mathbf{x})$, the anomaly score can be noisy for a single forward pass. In practice, the reconstruction error is averaged over K samples (e.g. $K = 10$):

$$s_{\text{VAE}}(\mathbf{x}) \approx \frac{1}{K} \sum_{k=1}^K \|\mathbf{x} - g_{\phi}(\mathbf{z}^{(k)})\|^2 + \text{KL}(q_{\theta}(\mathbf{z} | \mathbf{x}) \| p(\mathbf{z})).$$

7.7 Reconstruction Probability

An alternative to the ELBO-based score is the *reconstruction probability* approach of An and Cho [6], which provides a probabilistic anomaly score.

Definition 7.3: Reconstruction Probability

Let $p_{\phi}(\mathbf{x} | \mathbf{z}) = \mathcal{N}(g_{\phi}(\mathbf{z}), \text{diag}(\gamma_{\phi}^2(\mathbf{z})))$ be a decoder with learned variance. The **reconstruction probability** of $\mathbf{x}$ is:

$$p_{\text{recon}}(\mathbf{x}) = \frac{1}{K} \sum_{k=1}^K p_{\phi}(\mathbf{x} | \mathbf{z}^{(k)}), \quad \mathbf{z}^{(k)} \sim q_{\theta}(\mathbf{z} | \mathbf{x}). \quad (13)$$

The anomaly score is then $s(\mathbf{x}) = -\log p_{\text{recon}}(\mathbf{x})$. Unlike the MSE-based score, this accounts for the learned *per-feature variance* $\gamma_{\phi}^2(\mathbf{z})$: a large reconstruction error in a low-variance dimension is penalised more heavily than the same error in a high-variance dimension.

Remark 7.2: Advantage of reconstruction probability

By learning the decoder variance, the model can assign different levels of “expected noise” to different features. This is particularly useful when some features are inherently noisier than others: the anomaly score will not be dominated by reconstruction errors in high-variance (uninformative) dimensions. The reconstruction probability naturally calibrates the anomaly score to the learned noise structure of the data.

7.8 The β -VAE and Disentanglement

Definition 7.4: β -VAE

The β -VAE [2] modifies the standard VAE objective by introducing a hyperparameter $\beta > 0$ that controls the relative weight of the KL regularisation:

$$\mathcal{L}_{\beta\text{-VAE}} = \mathbb{E}_{q_{\theta}(\mathbf{z}|\mathbf{x})}[-\log p_{\phi}(\mathbf{x}|\mathbf{z})] + \beta \cdot \text{KL}(q_{\theta}(\mathbf{z}|\mathbf{x})\|p(\mathbf{z})). \quad (14)$$

Proposition 7.2: Effect of β on the latent representation

The parameter β controls a trade-off between reconstruction fidelity and latent space structure:

1. When $\beta < 1$, the model prioritises reconstruction quality. The latent space is less regularised and may exhibit holes or discontinuities.
2. When $\beta = 1$, the standard VAE (ELBO) objective is recovered.
3. When $\beta > 1$, stronger regularisation pressure encourages each latent dimension to capture an independent factor of variation (**disentanglement**). The latent space becomes more compact and structured, but reconstruction quality degrades.

For anomaly detection, $\beta > 1$ can improve separation between normal and anomalous samples by forcing the model to compress normal patterns more tightly around the prior, making anomalous deviations more conspicuous.

8 Latent Space Geometry and Interpretation

The structure of the latent space differs fundamentally between vanilla autoencoders and VAEs, with important consequences for anomaly detection.

8.1 Vanilla AE Latent Space

In a deterministic autoencoder, the latent space is **unregularised**. The encoder is free to place latent codes anywhere in $\mathbb{R}^{d_z}$, which can result in a fragmented, non-continuous latent space. Normal and anomalous points may end up interleaved in unpredictable ways, and the latent space offers no meaningful notion of density.

8.2 VAE Latent Space

The KL regularisation in the VAE encourages the aggregate posterior $\bar{q}(\mathbf{z}) = \frac{1}{N} \sum_i q_{\theta}(\mathbf{z}|\mathbf{x}_i)$ to approximate the standard normal prior, producing a latent space that is **smooth** and **continuous**. Points that are similar in input space tend to be nearby in latent space, and interpolation between latent codes produces semantically meaningful outputs. For anomaly detection, this means anomalous inputs are pushed to low-density regions of the prior, providing an additional geometric signal.

Remark 8.1: Posterior collapse

A known failure mode of VAEs is **posterior collapse**, where the encoder learns to ignore the input and map all observations to the prior, i.e. $q_{\theta}(\mathbf{z}|\mathbf{x}) \approx p(\mathbf{z})$ for all $\mathbf{x}$. In this regime, the KL term is near zero, but the reconstruction is poor and non-discriminative. Techniques to mitigate this include:

- **KL annealing**: gradually increasing β from 0 to 1 during training.
- **Free-bits**: imposing a minimum KL contribution per latent dimension.
- **δ -VAE**: choosing a decoder family that prevents collapse.

9 Comparison: Vanilla AE vs VAE

Table 3 summarises the key differences.

Table 3: Comparison of the vanilla autoencoder and the variational autoencoder for anomaly detection.

Property	Vanilla AE	VAE
Latent space	Deterministic point $\mathbf{z}$	Stochastic $\mathbf{z} \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\sigma}^2)$
Training objective	MSE reconstruction loss	ELBO = MSE + KL
Anomaly score	Reconstruction error	Reconstruction + KL divergence
Latent structure	Unregularised	Smooth, continuous (regularised)
Generative ability	No	Yes (sample from prior)
Robustness	Can overfit noise	More robust via KL regularisation
Computational cost	Lower	Higher (sampling, KL computation)

10 Limitations and Extensions

10.1 Limitations of Autoencoder-Based Anomaly Detection

- L1. Threshold sensitivity.** The performance of any reconstruction-based method depends critically on the choice of τ . Without labelled anomalies for validation, the threshold must be set heuristically, which can lead to suboptimal performance.
- L2. Partial anomalies.** If an anomalous input differs from normal data in only a few dimensions while matching in the majority, the averaged reconstruction error may remain below the threshold. Feature-wise anomaly scores can mitigate this issue.
- L3. No explicit density model.** The vanilla autoencoder provides an anomaly *score* but not a probability. It does not estimate $p(\mathbf{x})$ or provide calibrated confidence values.
- L4. Curse of dimensionality.** In very high-dimensional spaces, the MSE averages over many features, potentially diluting the signal from a few anomalous dimensions.
- L5. Training data contamination.** If anomalies are present in the training set, the autoencoder may learn to reconstruct them, reducing its discriminative ability.

10.2 Extensions

- **β -VAE [2]:** increasing the weight β on the KL divergence produces a more disentangled and compact latent representation, which can improve anomaly separation at the cost of reconstruction quality.
- **Memory-Augmented Autoencoder (MemAE) [3]:** augments the bottleneck with a learnable memory module containing prototypical latent codes. During reconstruction, the model is forced to use convex combinations of these prototypes, making it harder to reconstruct out-of-distribution inputs.
- **Deep Autoencoding Gaussian Mixture Model (DAGMM) [4]:** combines an autoencoder with a Gaussian Mixture Model (GMM) fitted jointly on the latent codes and reconstruction errors, providing an explicit density estimate for anomaly scoring.
- **LSTM-Autoencoder [5]:** replaces the dense encoder/decoder with recurrent (LSTM) layers for sequence data, enabling temporal anomaly detection in time series.

- **Convolutional Autoencoder:** uses convolutional layers for spatially structured data (images, spectrograms), where the bottleneck is a compressed spatial feature map rather than a flat vector.

References

References

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